

# Annu Tiwari

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LinkedIn | GitHub | LeetCode | Portfolio

## SUMMARY

Computer Science undergraduate with hands-on experience in full-stack development, backend APIs, and AI/ML applications. Built practical software projects and data-driven systems, with IEEE-published research in EEG and brain-computer interface systems.

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, TypeScript, JavaScript, SQL

**Machine Learning & AI:** Supervised Learning, Feature Engineering, Data Preprocessing, Model Evaluation, Explainable AI (SHAP), Scikit-learn, NumPy, Pandas, LLM Integration, Prompt Engineering

**Web & Backend:** Next.js, React.js, Node.js, Express.js, FastAPI, REST APIs, Tailwind CSS, JWT Authentication

**Databases & ORM:** MongoDB, PostgreSQL, Prisma

**Tools & Platforms:** Git, GitHub, VS Code, Postman, Jupyter Notebook, Google Colab, Overleaf, Chrome Extension Development

**Core CS & Security:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

## PROJECTS

### INORA - Multi-Channel EEG-Based Control

Dec 2025 - Jan 2026

**Technologies:** AD620, TL072, ADS1115, ESP32, Python, NumPy

- Built a low-cost 3-channel EEG acquisition system using AD620, ADS1115, and ESP32 with wireless streaming at **83 Hz/channel**.
- Developed a Python pipeline for EEG acquisition, filtering, and DRL-based noise reduction, and validated the system on **20 volunteers**.
- Demonstrated real-time EEG-based Pong game control and recorded physiologically valid EEG signals in the **15-80 uV** range using hardware costing approximately **INR 3,146**.

### BrainScore - EEG-Based Depression Detection

Mar 2025 - Nov 2025

**Technologies:** Python, Pandas, Scikit-learn, LightGBM, SHAP, Google Colab

- Processed EEG recordings from **58 subjects** using **20-channel** signals and extracted **18** statistical, frequency, and entropy-based biomarkers.
- Built and evaluated six machine learning models, using **SHAP** to interpret feature importance and improve model explainability.
- Achieved **95.28%** classification accuracy with a **LightGBM** model for automated depression detection.

### ApplyPilot - AI-Powered Job Application Platform

Jul 2026 - Present

**Technologies:** Next.js, TypeScript, React.js, FastAPI, Python, MongoDB, Groq API, JWT

- Engineered a **full-stack AI job application platform** providing automated ATS analysis, resume optimization, personalized cover letters, job matching, and interview preparation.
- Developed secure **FastAPI REST APIs** with **MongoDB** for authentication, resume management, job applications, AI workflows, and usage management.
- Built a **browser extension** that extracts job information from career websites and connects it with the user's selected resume for streamlined job analysis.

## EXPERIENCE

### AI Engineer Intern

Dec 2025 - Feb 2026

Infosys Springboard

Remote

- Cleaned and preprocessed cricket match datasets and engineered performance-based features.
- Trained machine learning models to predict player runs and wickets for upcoming matches.
- Built an interactive dashboard to visualize predictions and player performance trends.

### Machine Learning & Data Science Intern

May 2025 - Jul 2025

Svaarogyam Medical Devices Pvt. Ltd.

Remote

- Worked on healthcare machine learning tasks including data cleaning, preprocessing, feature extraction, and predictive modeling.

- Built end-to-end machine learning pipelines for healthcare use cases.
- Collaborated with a remote team to deliver project milestones in the medical devices domain.

## EDUCATION

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**IILM University, Greater Noida**

B.Tech in Computer Science and Engineering

**2023 - 2027**

**CGPA: 8.74/10**

## ACHIEVEMENTS / CERTIFICATIONS

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- Research article accepted in IEEE Transactions on Medical Robotics and Bionics (T-MRB): SHAP-based Study: XAI-Guided Optimisation of Spatial-Temporal Bio-Neuro Marker EEG for Depression
- Research Paper accepted at IEEE Sponsored ETNCC 2026 for INORA: A Low-Cost 3-Channel EEG Acquisition System with Driven-Right-Leg Noise Suppression for BCI Prototyping
- Hackathon Winner (1st Rank) - Secured 1st place among 1,004 participants (118 teams)
- Secured 1st place in the Call for Paper event at SRM Institute of Science and Technology
- Completed Infosys Springboard Pragati: Path to Future - Cohort 6
- Solved 300+ Data Structures & Algorithms problems on LeetCode